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1. Which of the following processes provide evidence of the particulate nature of matter?

- I. Capillarity
- II. Diffusion
- III. Filtration
- IV. Osmosis

- (A) I and III only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

2. 'Mass number' is the number of

- (A) neutrons plus protons
- (B) neutrons minus protons
- (C) electrons plus neutrons
- (D) electrons plus protons

3. Which of the following elements has 7 electrons in its outer shell?

- (A) Hydrogen
- (B) Oxygen
- (C) Nitrogen
- (D) Chlorine

4. An atom that has 20 electrons when it ionizes has the same electronic configuration as

- (A) argon
- (B) calcium
- (C) neon
- (D) sodium

5. $^{24}_{11}\text{Na}$ and $^{23}_{11}\text{Na}$ are isotopes because they

- (A) conduct electricity
- (B) react vigorously with water
- (C) have the same number of protons
- (D) have the same relative atomic mass

6. Sulfur and oxygen are in the same group of the periodic table because

- (A) they can react with each other
- (B) they can form covalent compounds
- (C) they have the same number of electrons in their outer shell
- (D) the atomic number of sulfur is 16 and the relative atomic mass of oxygen is 16

Item 7 refers to the following quantities of atoms.

- I. 12 g of carbon
- II. 16 g of sulfur
- III. 23 g of sodium
- IV. 12 g of magnesium

(Relative mass number: C = 12; S = 32; Mg = 24; Na = 23)

7. Which of the following pairs represents equal quantities of atoms?

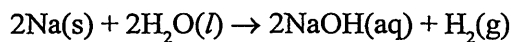
- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

8. An ion with a single negative charge may be converted into a neutral atom by

- (A) losing a neutron
- (B) gaining a proton
- (C) losing an electron
- (D) gaining an electron

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9. Sodium reacts with water according to the equation



(Molar volume of gas = 24 litres at rtp).

The number of litres of hydrogen (at rtp) liberated when 0.1 mole of sodium reacts with excess water is

- (A) 1.2
(B) 2.4
(C) 12.0
(D) 24.0

10. A new element, R, shows chemical properties similar to the element sodium. What is the correct formula of a compound formed with R and sulfur?

- (A) RS_2
(B) RS
(C) R_2S
(D) R_2S_2

11. Barium is below calcium in Group II of the Periodic Table. When BOTH metals react with water, the MAIN differences in observation would be the rate of reaction and the

- (A) type of product
(B) type of gas evolved
(C) reaction with litmus
(D) solubility of product

Item 12 refers to the following information.

Element	Atomic Number
I.	3
II.	6
III.	17
IV.	18

12. Which TWO of the elements above when combined with each other form MAINLY ionic compounds?

- (A) I and II only
(B) I and III only
(C) II and IV only
(D) III and IV only

Items 13–14 refer to the following terms.

- (A) Giant ionic structure
(B) Simple molecular crystal
(C) Macromolecular structure
(D) Giant covalent structure

Match EACH substance below with one of the terms above, each of which may be used more than once, once or not at all.

13. Calcium chloride

14. Starch

15. A separating funnel can be used to separate a mixture of

- (A) water and ethanol
(B) water and kerosene
(C) water and solid sodium chloride
(D) kerosene and solid sodium chloride

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16. Which of the following substances, when added to pure water, SIGNIFICANTLY increases the water's conductivity?
- (A) Graphite
(B) Iron(II) hydroxide
(C) Copper(II) hydroxide
(D) Sodium chloride
17. A solution having a pH of 1 will
- (A) neutralize a solution of pH 4
(B) react with zinc metal to produce hydrogen
(C) react with zinc metal to produce a solution of pH 10
(D) react with hydrochloric acid to produce a salt and water
18. Which of the following halogens is a liquid at room temperature?
- (A) Chlorine
(B) Bromine
(C) Fluorine
(D) Iodine
19. A 'weak acid' is BEST described as one that yields a
- (A) low reading on the pH meter
(B) low concentration of acid in aqueous solution
(C) high concentration of hydrogen ions in solution
(D) low concentration of hydrogen ions in aqueous solution
20. Which of the following methods may be used in the preparation of barium sulfate in the laboratory?
- (A) Distillation
(B) Evaporation
(C) Precipitation
(D) Crystallization
21. Which TWO of the following statements describe acid salts?
- I. Their anions cannot dissociate to give hydrogen ions.
II. They form a normal salt when all replaceable hydrogen of the acid is removed.
III. Their anions can dissociate to give hydrogen ions.
IV. They are not capable of generating a normal salt.
- (A) I and III
(B) I and IV
(C) II and III
(D) III and IV
22. In which of the following reactions is sulfur dioxide acting as an oxidizing agent?
- (A) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$
(B) $\text{SO}_2(\text{g}) + 2\text{H}_2\text{S}(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + 3\text{S}(\text{s})$
(C) $\text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_3(\text{aq})$
(D) $\text{SO}_2(\text{g}) + 2\text{HNO}_3(\text{aq}) \rightarrow \text{H}_2\text{SO}_4(\text{aq}) + 2\text{NO}_2(\text{g})$
23. In which of the following compounds does hydrogen have a negative oxidation number?
- (A) CH_4
(B) H_2O_2
(C) NH_3
(D) NaH

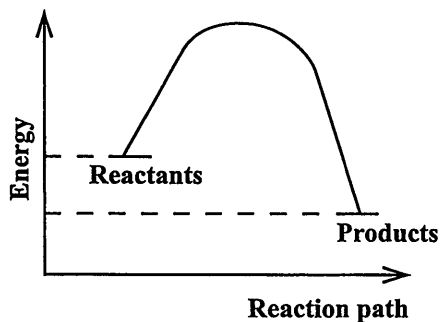
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24. When an exothermic reaction occurs, energy is
- (A) absorbed from the environment
 - (B) released to the environment
 - (C) absorbed from the reactants
 - (D) released from the products
25. Which of the following observations is expected with the electrolysis of concentrated sodium chloride solution using graphite electrodes?
- (A) Chlorine is evolved at the anode.
 - (B) The pH of the solution decreases.
 - (C) Oxygen is evolved at the anode.
 - (D) Sodium is discharged at the cathode.
26. A solution, X, is added to acidified aqueous potassium iodide which turns brown. Addition of starch to the solution results in a dark blue colouration. It is possible that solution X contains
- (A) Fe^{2+} ions
 - (B) SO_3^{2-} ions
 - (C) a reducing agent
 - (D) an oxidizing agent
27. In which of the following substances will electrolysis NOT occur when an electrical current is passed through it?
- (A) Dilute sulfuric acid
 - (B) Solid sodium chloride
 - (C) Molten sodium chloride
 - (D) Sodium hydroxide solution
28. Which of the following factors will usually increase the rate of catalytic decomposition of hydrogen peroxide?
- (A) Increasing the temperature
 - (B) Applying a greater external pressure
 - (C) Using a larger volume of hydrogen peroxide
 - (D) Using the catalyst in lumps rather than in powdered form
29. Which of the following substances is an amphoteric oxide?
- (A) Magnesium oxide
 - (B) Aluminium oxide
 - (C) Silicon dioxide
 - (D) Sulfur dioxide

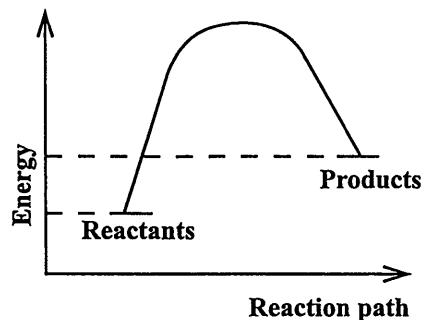
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30. Which of the following diagrams BEST illustrates the course of an exothermic reaction?

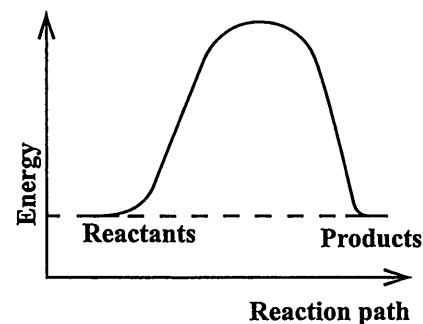
(A)



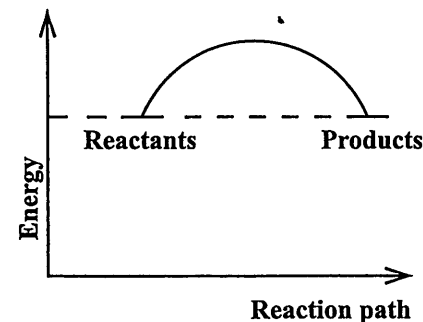
(B)



(C)



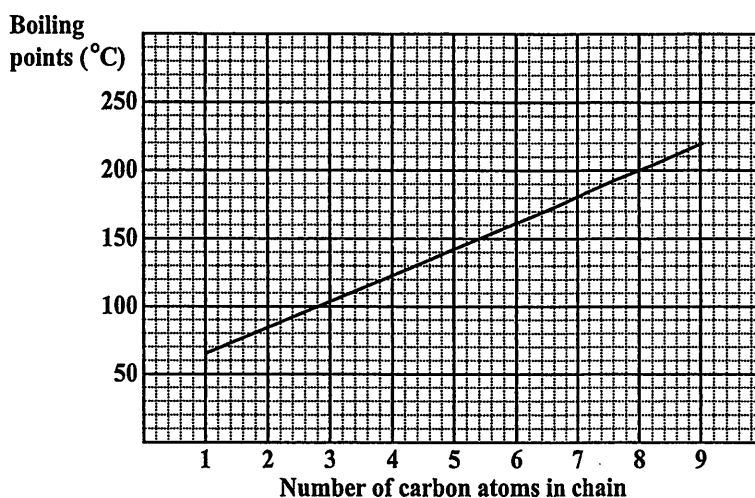
(D)



31. ALL members of a homologous series have similar

- (A) densities
- (B) boiling points
- (C) chemical properties
- (D) physical properties

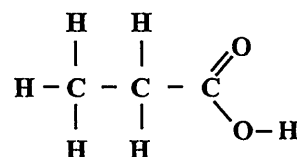
Item 32 refers to the following graph which shows the boiling points of some straight-chain alcohols.



32. Which of the following molecular formulae represents the alcohol whose boiling point is 180 °C?

- (A) $\text{CH}_3(\text{CH}_2)_3\text{OH}$
- (B) $\text{CH}_3(\text{CH}_2)_4\text{OH}$
- (C) $\text{CH}_3(\text{CH}_2)_5\text{OH}$
- (D) $\text{CH}_3(\text{CH}_2)_6\text{OH}$

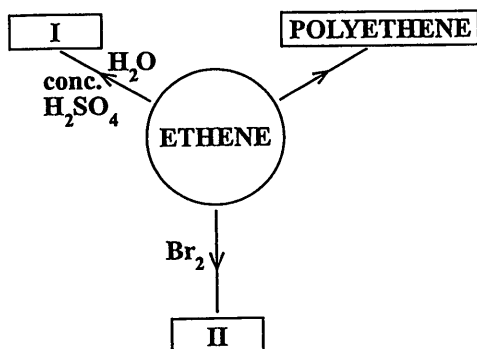
33. To which homologous series does the following structure belong?



- (A) Acids
- (B) Alkanes
- (C) Alkenes
- (D) Alcohols

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Items 34–35 refer to the following chart showing the conversion of ethene to different products.



34. The product I is

- (A) ethane
- (B) ethanol
- (C) ethanoic acid
- (D) ethyl ethanoate

35. The product, polyethene, is a

- (A) gas
- (B) liquid
- (C) solid
- (D) solution

36. Which of the following reactions occurs between propene and bromine?

- (A) Addition
- (B) Condensation
- (C) Substitution
- (D) Precipitation

Items 37–38 refer to the following equation.



37. The forward reaction is known as

- (A) synthesis
- (B) hydrolysis
- (C) esterification
- (D) polymerization

38. The reverse reaction is known as

- (A) synthesis
- (B) hydrolysis
- (C) esterification
- (D) polymerization

39. When large alkane molecules are cracked, the products are a mixture of

- (A) small alkane molecules
- (B) small alkene molecules
- (C) large and small alkane molecules
- (D) small alkane and alkene molecules

40. Which of the following reagent(s) can be used for converting ethanol to ethanoic acid?

- (A) Cl_2
- (B) H_2O
- (C) $\text{H}^+/\text{K}_2\text{Cr}_2\text{O}_7$
- (D) Steam, H_3PO_4 catalyst

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41. The fermentation of sugars, using glucose as the substrate, can be represented by the equation
- (A) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$
(B) $C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$
(C) $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
(D) $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
42. Glucose is converted to starch or cellulose by
- (A) dehydrogenation
(B) oxidation and reduction
(C) addition polymerization
(D) condensation polymerization
43. Which of the following features belong to metals?
- I. Form solid chlorides
II. Form basic oxides
III. Conduct electricity when molten
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III
44. Which of the following processes produces a gas which is different from the others?
- (A) Respiration in animals
(B) Photosynthesis in plants
(C) Heating calcium carbonate
(D) Oxidation of carbon monoxide
45. Which of the following metals will NOT displace hydrogen from dilute hydrochloric acid?
- (A) Zinc
(B) Iron
(C) Copper
(D) Aluminium
46. Which of the following methods is used for the extraction of aluminium?
- (A) Electrolysis of its molten oxide
(B) Electrolysis of its aqueous chloride
(C) Reduction of its oxide using coke
(D) Reduction of its oxide using carbon monoxide
47. Which of the following statements about sulfur are true?
- I. It is used in the vulcanization of rubber.
II. It is used in the extraction of iron from iron ore.
III. It is used in the manufacture of explosives.
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III
48. Which of the following substances conducts an electric current but remains chemically unchanged?
- (A) Aqueous copper(II) sulfate
(B) Copper
(C) Sulfur
(D) Sodium chloride

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49. Which of the following gases is NOT considered to be toxic?

- (A) Carbon dioxide
- (B) Sulfur dioxide
- (C) Nitrogen dioxide
- (D) Carbon monoxide

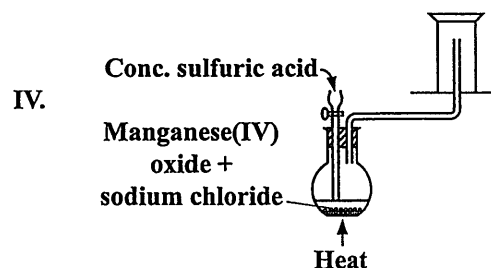
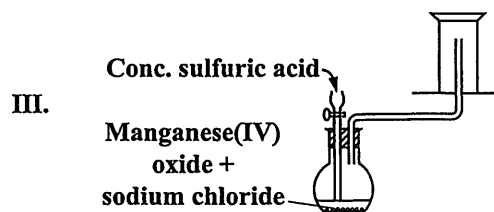
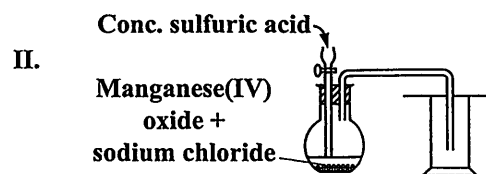
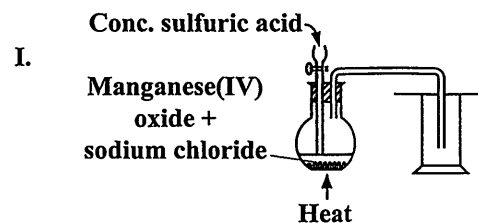
50. The compound which has the formula $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_2\text{CH}_3$ is

- (A) an alkene
- (B) an unsaturated acid
- (C) a carboxylic acid
- (D) a polymer

51. When copper(II) carbonate is heated alone in a dry test tube, a gas is evolved and a black residue is formed. This gas is expected to

- (A) turn red litmus blue
- (B) relight a glowing splint
- (C) decolourize acidified aqueous potassium manganate(VII)
- (D) form a white precipitate with aqueous calcium hydroxide

Item 52 refers to Experiments I-IV shown in the following diagrams.



52. In which experiment would the amount of chlorine collected be GREATEST?

- (A) I
- (B) II
- (C) III
- (D) IV

53. Which of the following observations is expected when aqueous silver nitrate is added to aqueous potassium chloride?

- (A) A pungent gas is evolved.
- (B) A blue precipitate appears.
- (C) A white precipitate appears.
- (D) A brown precipitate appears.

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Items 54–55 refer to the following metals.

- (A) Al
- (B) Cu
- (C) Fe
- (D) Na

A particular metal may be selected more than once, once or not at all.

54. Which metal reacts vigorously with water to produce a strongly alkaline solution?
55. Which metal has an alloy which is used in the manufacture of aircraft?
56. A gas, Z, has physical and chemical properties similar to ammonia. The BEST method for collecting a pure, dry sample of Z would be to pass it through
- (A) water and collect by downward delivery
 - (B) calcium oxide and collect by upward delivery
 - (C) calcium chloride and collect by downward delivery
 - (D) concentrated sulfuric acid and collect by upward delivery
57. The main substances responsible for 'acid rain' are
- (A) chlorofluorocarbons
 - (B) lead compounds in exhaust fumes
 - (C) carbon dioxide and carbon monoxide
 - (D) sulfur dioxide and sulfur trioxide
58. Although aluminium is more reactive than iron, aluminium becomes corrosion resistant when left exposed to the atmosphere while iron corrodes. This is because the
- (A) aluminium has reacted completely
 - (B) iron oxide formed is more reactive than the aluminium oxide formed
 - (C) oxide film on aluminium protects the metal from further corrosion but the oxide film on iron does not
 - (D) iron oxide reacts with the iron below the surface but the aluminium oxide does not react with the aluminium below the surface
59. Which of the following substances is an important constituent of chlorophyll?
- (A) Iron
 - (B) Copper
 - (C) Calcium
 - (D) Magnesium
60. Which of the following gases will give a positive test with dry cobalt chloride paper?
- (A) CO₂
 - (B) O₂
 - (C) H₂
 - (D) H₂O

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.